

TANVIR AHMED

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I am a final-year Computer Science and Engineering student at North South University, working on multimodal machine learning: which inputs actually need an expensive model, and when a model should say that it does not know. One accepted paper, six more under review, and three months as a research assistant at the NSU Trimodal AI Lab. Alongside the research I build full-stack systems, and I have spent time on quantum machine learning and on inference for microcontrollers.

RESEARCH INTERESTS

- Efficient multimodal inference
- Computer vision and video understanding
- Calibration, abstention and uncertainty
- Agentic systems
- Quantum machine learning
- Explainable AI
- Edge and embedded machine learning
- Full-stack system architecture

EDUCATION

Bachelor of Science in Computer Science and Engineering

Aug 2022 – Dec 2026

North South University, Dhaka, Bangladesh. Department of Electrical and Computer Engineering. Expected graduation December 2026.

CGPA: 2.99 / 4.00

Higher Secondary Certificate (HSC), Science

Sep 2019 – Dec 2021

Milestone College, Dhaka. GPA: 5.00 / 5.00

EXPERIENCE

Research Assistant, Trimodal AI Lab

Mar 2026 – May 2026

North South University, Dhaka, Bangladesh. Contract, on-site, academic research.

- Worked on multimodal and agentic AI systems under faculty supervision: reading the literature, testing ideas and building prototypes.
- Ran experiments with other lab members and wrote up the results, including what they did not show.
- Part of the lab since 2025, through its reading group and joint projects.

PUBLICATIONS

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

Accepted

Tanvir Ahmed. *LM-KBC 2026 shared task (5th edition), Automated Knowledge Base Construction (AKBC) at EMNLP 2026*.

- Set-valued knowledge base construction with an open-weight model of at most 32B parameters, no retrieval and no fine-tuning. Deciding *when to answer* is treated as a first-class decision, and the system beats the official baseline on five of the six relations.

Code: github.com/minhaz-42/lm-kbc-2026

SoundSentry: A Low-Cost Edge-AI Acoustic Event Detection and Direction-Finding System on STM32

Under review

Tanvir Ahmed, Sanzana Afrin Tonny, Husain Kabir Sifat. *International Conference on Electrical and Computer Engineering (ICECE) 2026, Dhaka*.

- STM32F103 prototype with a three-microphone array, ADC-DMA acquisition, on-device classification over 11 acoustic features, servo-based direction response and SD-card logging.

Code: github.com/minhaz-42/SoundSentry

SimCert: A Classical-Simulability Audit of Trained QNLP Text Models

Under review

Submitted to QTML 2026.

- Re-simulates trained quantum-NLP circuits under matrix-product-state truncation, and ablates entanglement, to test whether the quantum component is load-bearing.

Technologies: Python, PennyLane, lambeq, OpenQASM, Hydra

Code: github.com/minhaz-42/simcert-qlnp

Observing Is Not Free: The Decision Value of Visual Observations in Vision-Language Agents

Under review

- Asks whether an agent’s observation-selection criterion ranks observations the way their realised effect on the answer does, measured with a counterfactual table of the model’s answer before and after every candidate observation. Decision value turns out not to be predictable from crop geometry.

Technologies: Python, Vision-Language Models

Knowing What You No Longer Know: Calibrated Belief Validity for Persistent World Memory

Under review

- Separates whether a belief is still true from how confident the model was when it formed it, and builds a benchmark of elapsed-time-matched pairs over real and simulated changing rooms to score the difference.

Technologies: Python, PyTorch, HD-EPIC, AI2-THOR, ProcTHOR

Sequential Video Belief

Under review

- Tests whether a video-language model’s own stated answer impairs its ability to revise that answer when later visual evidence contradicts it, with the go or no-go rule fixed before analysis.

Technologies: Python, Vision-Language Models

Uncertainty-Guided Language-Grounded 3D Scene Graphs from Sparse Views

Under review

- Frozen geometry and vision-language models with a lightweight graph reasoning module, for object-level 3D segmentation and spatial relations from only a few casual images.

Technologies: Python, PyTorch, VGGT, SAM 2, CLIP, DINOv2

RESEARCH PROJECTS

Frame-Aware Multi-Model Scheduling for Efficient Video Captioning

Senior Design, 2026

Routes video captioning between a 4B and a 27B vision-language model so the large model is spent only where the footage is hard. Lead author of four.

- Frames are stitched into mosaics, encoded by a frozen CLIP ViT-B/32 and scored for visual complexity by a small graph attention network trained without human labels.
- Answers a 19-clip benchmark in fewer requests than either model alone, while scoring above the small model on every judged axis.

Technologies: Python, PyTorch, PyTorch Geometric, CLIP, Django, Gemma 3, Qwen3-VL

Quantum-Enhanced Driver Distraction Detection

Directed Research, 2026

A hybrid classical–quantum pipeline with uncertainty-aware risk scoring, tested under a protocol where only the classification head differs.

- The variational quantum head reaches accuracy parity with its matched classical control and cuts expected calibration error by 37% relative.
- The report states what the evidence does not support, including that the calibration gain cannot be attributed to the circuit.

Technologies: Python, PyTorch, PennyLane, EfficientNet, scikit-learn

Bangla Mental Health Conversational Agent

Ongoing

Mental health chatbot for Bangla-speaking users, built on fine-tuned Gemma-3-Bangla and Qwen-2.5-0.5B-Bangla models, with voice input and output, mood tracking, conversation history and a multi-language interface.

Technologies: Python, Django, PyTorch, Transformers, OAuth, Apple MPS, SQLite

TECHNICAL PROJECTS

Pakna Council: AI Council Platform

2026

Local-first web app that runs several LLMs in parallel on one prompt through Ollama, with model comparison and vision workflows.

Technologies: Python, Django, React, Docker, Ollama, WebSocket

Links: [Source](#) · [Live site](#)

Nestledown: 3D Exploration Game	2026
Cozy 3D exploration game built solo in Godot 4, with every model, texture, animation and sound generated by a script and no third-party asset in the repository. A parallel build of the same world runs on a custom C++20 engine.	
Technologies: Godot 4, Blender, C++20, SDL3, CMake	
Links: Announcement trailer	
SHAND: Explainable Assumption Detector	2026
Rule-based system for finding assumptions in English text, with visual analytics and export.	
Technologies: Python, NLP, HTML, JavaScript, D3.js	
Links: Source · Live site	
ISKA: Privacy-First Document Analysis Assistant	2026
Document analysis assistant with a Chrome extension for use while reading.	
Technologies: TypeScript, Chrome extension APIs	
Links: Source · Live site	
SUMBA: 3D Sign Language Translation	2026
Real-time sign-to-text translation using computer vision, with mobile integration and offline use.	
Technologies: Django, MediaPipe, Transformers, React Native	
Links: Source	
Entangled Fusion Quantum Circuits for Vision–Language Alignment	2026
Experimental multimodal platform that combines vision analysis with quantum-inspired fusion and natural language reasoning.	
Technologies: Python, Django, PennyLane, Ollama, OpenCV, REST APIs	
Links: Source	
FitWell: AI Nutrition Assistant	2025
AI-powered meal planning, BMI tracking and health insights.	
Technologies: Django, Python, Qwen, SQLite, Chart.js	
Links: Source	
DG Healthcare Management System	2024
Hospital management platform with appointments, billing and role-based access.	
Technologies: HTML5, CSS3, PHP, MySQL, JavaScript	
Links: Source	
CSE215 Java Application	2024
Object-oriented Java application with a modular MVC structure and unit tests.	
Technologies: Java, JavaFX, MySQL, JUnit	
Links: Source	

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++, PHP, SQL, HTML, CSS, LaTeX

Machine Learning and AI: Deep learning, computer vision, NLP, LLMs, Transformers, vision-language models, fine-tuning, calibration and abstention, quantum machine learning

Frameworks and Libraries: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Transformers, OpenCV, CLIP, MediaPipe, PennyLane, lambeq, Hydra

Software and Web: Django, React, Node.js, JavaFX, REST APIs, WebSocket, Docker, CI/CD, OOP, design patterns, Agile/Scrum, unit testing

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Embedded and Graphics: STM32, embedded C, ADC-DMA acquisition, Godot 4, Blender, SDL3, CMake

Tools: Git, Linux, VS Code, Jupyter, LaTeX, Ollama, Apple MPS

EXTRACURRICULAR ACTIVITIES

IEEE NSU Student Branch, General Member

2023 – Present

Technical workshops, seminars and community outreach programmes.

REFERENCES

Md Tanvir Ahmad Palash

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